

AWS SQS (Simple Queue Service)

1. Objective

To understand AWS Simple Queue Service (SQS), create a queue, send and receive messages, and learn how SQS enables asynchronous communication between applications.

2. What is AWS SQS?

AWS Simple Queue Service (SQS) is a fully managed message queuing service that enables different components of an application to communicate reliably without being directly connected.

Instead of one application directly calling another, it places a message in a queue. Another application retrieves and processes the message whenever it is ready.

Definition

AWS SQS is a fully managed message queuing service that allows applications to exchange messages asynchronously, improving scalability, reliability, and fault tolerance.

3. Why Do We Need SQS?

Imagine an e-commerce website.

When a customer places an order, multiple tasks need to happen:

- Save the order
- Send a confirmation email
- Update inventory
- Generate an invoice
- Notify the delivery team

If all of these happen at once, the application may become slow.

Instead, the application sends messages to an SQS queue. Different services process those messages independently.

4. Features of AWS SQS

- Fully managed service
- Highly scalable
- Reliable message delivery
- Supports asynchronous communication
- Secure with IAM

- Integrates with Lambda, EC2, SNS, EventBridge, and more
 - No server management
-

5. Types of SQS Queues

Standard Queue

Unlimited throughput

Best-effort ordering

At-least-once delivery

Default queue type

FIFO Queue

Limited throughput

Strict message ordering

Exactly-once processing

Used when order matters

Examples

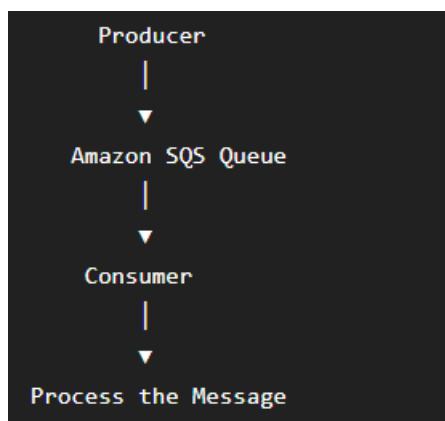
Standard Queue

- Website notifications
- Log processing
- Background tasks

FIFO Queue

- Banking transactions
 - Payment processing
 - Stock trading
-

6. AWS SQS Architecture



E-Commerce Order Processing Using AWS SQS

Step 1: Create the Queue

1. Open **AWS Console**.
2. Search **Amazon SQS**.
3. Click **Create queue**.
4. Select **Standard Queue**.
5. Queue Name:

OrderQueue

6. Keep all settings as default.
7. Click **Create queue**.

Screenshot : OrderQueue created

The screenshot shows the AWS SQS console interface. At the top, a green notification bar states "Queue OrderQueue created successfully" with a close button. Below this, the breadcrumb navigation shows "Amazon SQS > Queues > OrderQueue". The main heading is "OrderQueue", followed by action buttons: "Edit", "Delete", "Purge", "Send and receive messages", and "Start DLQ redrive". A "Details" section is expanded, showing a table of queue properties.

Details Info		
Name OrderQueue	Type Standard	ARN arn:aws:sqs:us-east-1:299582146630:OrderQueue
Encryption Amazon SQS key (SSE-SQS)	URL https://sqs.us-east-1.amazonaws.com/299582146630/OrderQueue	Dead-letter queue -

Step 2: Open the Queue

Click **OrderQueue**.

Click **Send and receive messages**.

Step 3: Send Order Messages

Instead of just writing "Hello", we'll send a realistic order.

Message 1

Order ID:1001

Customer:HARI

Product:Laptop

Quantity:1

Price:65000

Click **Send message**.

Message 2

Order ID:1002

Customer: ABU

Product:Mobile

Quantity:1

Price:50000

Click **Send message**.

Message 3

Order ID:1003

Customer:Mahesh

Product:Headphones

Quantity:2

Price:3000

Click **Send message**.

Screenshot: Three messages sent

Send and receive messages

Use this page to send, retrieve and view messages, helping you experiment with various queue features.

Send message [Info](#)

✓ Your message has been sent and is ready to be received.

[View sent message details](#)



Message body

Enter the message to send to the queue.

Order ID:1003
Customer:Mahesh
Product:Headphones

Message group ID - optional, new [Info](#)

A group identifier for the message to allow fair processing across message groups in a standard queue.

Step 4: Poll the Queue

Click

Poll for messages

You'll see:

Screenshot: Messages received

Receive messages [Info](#)

Edit poll settings

Stop polling

Poll for messages

Messages available
3

Polling duration
30

Maximum message count
10

Polling progress
1 receives/second 50%

Messages (3)

View details

Delete

Q Search messages

< 1 > ⚙

☐	ID ▾	Sent ▲	Size ▾	Receive count ▾
☐	007c0978-2860-41a5-bc9e-4b43829b2976	2026-08-05T16:15+05:30	65 bytes	1
☐	89d2d2e8-9b92-4ac0-8d12-89e29e2945ef	2026-08-05T16:16+05:30	64 bytes	1
☐	9e7d921c-eae3-4afd-a5f2-0ef288e8235e	2026-08-05T16:16+05:30	70 bytes	1

Step 1: Open Lambda

Go to:

AWS Console → Lambda → OrderQueue

Step 2: Add Trigger

On the Lambda page,

Click

Add Trigger

(Usually at the top of the Function Overview section.)

Step 3: Select Source

Choose

Amazon SQS

Step 4: Select Queue

Choose your queue

OrderQueue

Add trigger

Trigger configuration [Info](#)

 **SQS**

aws event-source-mapping polling queue

SQS queue
Choose or enter the ARN of an SQS queue.



Event poller configuration

☒ **Activate trigger**

Select to activate the trigger now. Keep unchecked to create the trigger in a deactivated state for testing (recommended).

Step 5: Batch Size

Keep

10

(Default)

Step 6: Enable Trigger

Make sure

Enable Trigger

is checked.

Click

Add

You'll now see:

Amazon SQS



Order processor

Screenshot: SQS Trigger Added

OrderProcessor

Throttle Copy ARN Actions

✓ The trigger OrderQueue was successfully added to function OrderProcessor. X

▼ Function overview Info

Diagram Template

 **OrderProcessor**
 Layers (0)

 **SQS**

+ Add trigger

+ Add destination

Export to Infrastructure Composer Download

Function ARN
arn:aws:lambda:us-east-1:299582146:630:function:OrderProcessor

Description
-

Last modified
11 minutes ago

Error: when creating the trigger



The function execution role does not have permissions to call ReceiveMessage on SQS.

This means your **Lambda IAM Role** doesn't have permission to read messages from the SQS queue.

Step 1: Open your Lambda function

Go to:

Lambda → HelloLambda

Click the **Configuration** tab.

Step 2: Open the Execution Role

Click

Permissions

You'll see something like:

Execution role

HelloLambda-role-xxxxxxx

Click the **role name**.

It opens the IAM Role.

Step 3: Add SQS Permission

Click

Add permissions

↓

Attach policies

Search for:

AWSLambdaSQSQueueExecutionRole

Select it.

Click

Add permissions.

This policy allows Lambda to:

- Receive messages
- Delete messages
- Read queue attributes

The screenshot shows the AWS IAM console interface. At the top, there is a search bar with the text 'awslambdasq' and a dropdown menu set to 'All types'. Below the search bar, a table lists policies. The first policy, 'AWSLambdaSQSQueueExecutionRole', is selected. The policy details are shown below the table, including a description: 'Provides receive message, delete message, and read attribute access to SQS queues, and write permissions to CloudWatch logs.' The policy JSON is displayed in a code editor. At the bottom right, there are two buttons: 'Cancel' and 'Add permissions'.

Search: awslambdasq All types 1 match

Policy name	Type	Description
AWSLambdaSQSQueueExecutionRole	AWS managed	Provides receive message, delete message, and read attribute access to SQS queues, and write permissions to CloudWatch logs.

AWSLambdaSQSQueueExecutionRole [Copy JSON](#)

Provides receive message, delete message, and read attribute access to SQS queues, and write permissions to CloudWatch logs.

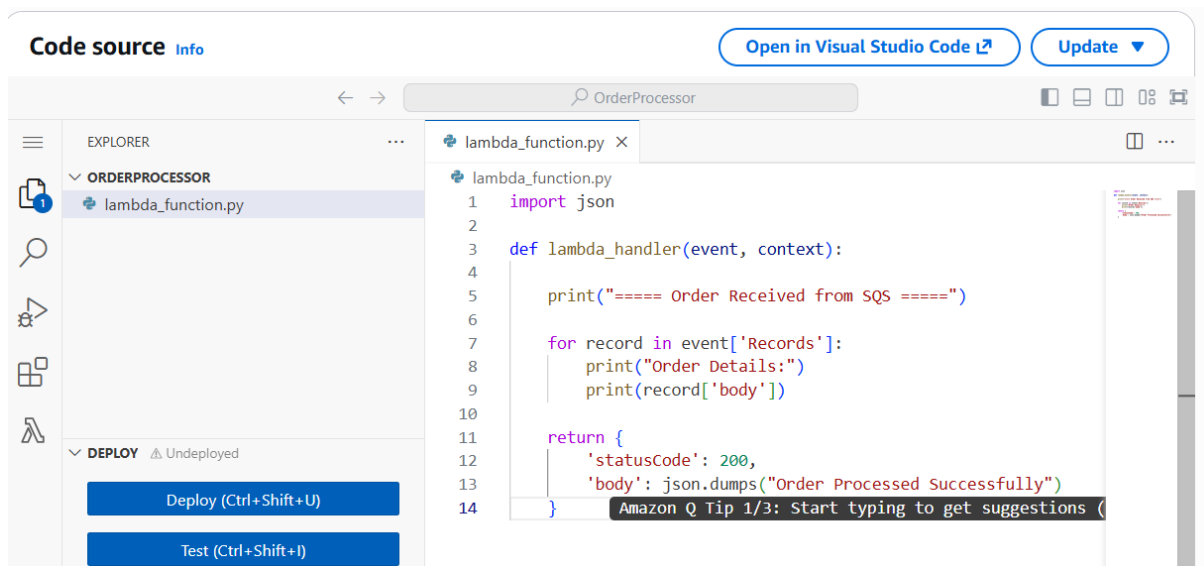
```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sqs:ReceiveMessage",
8         "sqs:DeleteMessage",
9         "sqs:GetQueueAttributes",
10        "logs:CreateLogGroup",
11        "logs:CreateLogStream",
12        "logs:PutLogEvents"
13      ],
14      "Resource": "*"
15    }
16  ]
17 }
```

[Cancel](#) [Add permissions](#)

Step 7: Replace Lambda Code

Go to **Code**.

Replace the existing code with:



Click **Deploy**.

Step 8: Send a Message to SQS

Go to

Amazon SQS → OrderQueue

Click

Send and receive messages

Message:

Order ID:1001

Customer:HARI

Product:Laptop

Quantity:1

Price:65000

Click

Send Message

Step 9: Verify Lambda Execution

Go back to

Lambda → Order processor

Open

Monitor

Click

View CloudWatch Logs

Open the latest Log Stream.

Screenshot: CloudWatch Logs

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms 0 0 0

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

▼ Logs

Log Management

Log Analytics New

Log Anomalies

Log events

Actions Start tailing Create metric filter

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

1m 1h UTC timezone Display

Timestamp	Message
2026-08-05T11:14:59.199Z	===== Order Received from SQS =====
2026-08-05T11:14:59.199Z	Order Details:
2026-08-05T11:14:59.199Z	Order ID:1001
2026-08-05T11:14:59.199Z	Customer:HARI
2026-08-05T11:14:59.199Z	Product:Laptop
2026-08-05T11:14:59.199Z	Quantity:1
2026-08-05T11:14:59.199Z	Price:65000
2026-08-05T11:14:59.200Z	END RequestId: 3c6a3ada-b571-5f5d-a448-5837a8039647

Step 10: Delete Messages

Select all messages.

Click

Delete

Queue becomes empty.

Screenshot: Queue empty

Receive messages Info

Edit poll settings Stop polling Poll for messages

Messages available 0

Polling duration 30

Maximum message count 10

Polling progress Success 0.1 receives/second

Messages (3)

View details Delete

Search messages

< 1 >

☒	ID	Sent	Size	Receive count
☒	007c0978-2860-41a5-bc9e-4b43829b2976	2026-08-05T16:15+05:30	65 bytes	2
☒	89d2d2e8-9b92-4ac0-8d12-89e29e2945ef	2026-08-05T16:16+05:30	64 bytes	1
☒	9e7d921c-eae3-4afd-a5f2-0ef288e8235e	2026-08-05T16:16+05:30	70 bytes	1